CONTENTS

MATHEMATICS

CHAPTER	TITLE	Page No.	Month
1	**Applications of Matrices and Determinants**	**1**	**Jun**
1.1	Introduction	1	
1.2	Inverse of a Non-Singular Square Matrix	2	
1.3	Elementary Transformations of a Matrix	16	
1.4	Applications of Matrices: Solving System of Linear Equations	27	
1.5	Applications of Matrices: Consistency of system of linear equations by rank method	38	
2	**Complex Numbers**	**52**	**Jul**
2.1	Introduction to Complex Numbers	52	
2.2	Complex Numbers	54	
2.3	Basic Algebraic Properties of Complex Numbers	58	
2.4	Conjugate of a Complex Number	60	
2.5	Modulus of a Complex Number	66	
2.6	Geometry and Locus of Complex Numbers	73	
2.7	Polar and Euler form of a Complex Number	75	
2.8	de Moivre's Theorem and its Applications	83	
3	**Theory of Equations**	**97**	**Jul**
3.1	Introduction	97	
3.2	Basics of Polynomial Equations	99	
3.3	Vieta's Formulae and Formation of Polynomial Equations	100	
3.4	Nature of Roots and Nature of Coefficients of Polynomial Equations	107	
3.5	Applications of Polynomial Equation in Geometry	111	
3.6	Roots of Higher Degree Polynomial Equations	112	
3.7	Polynomials with Additional Information	113	
3.8	Polynomial Equations with no additional information	118	
3.9	Descartes Rule	124	
4	**Inverse Trigonometric Functions**	**129**	**Jul/Aug**
4.1	Introduction	129	
4.2	Some Fundamental Concepts	130	
4.3	Sine Function and Inverse Sine Function	133	

12th_Maths_EM_Vol1_FP.indd 4 16-12-2022 14:27:20